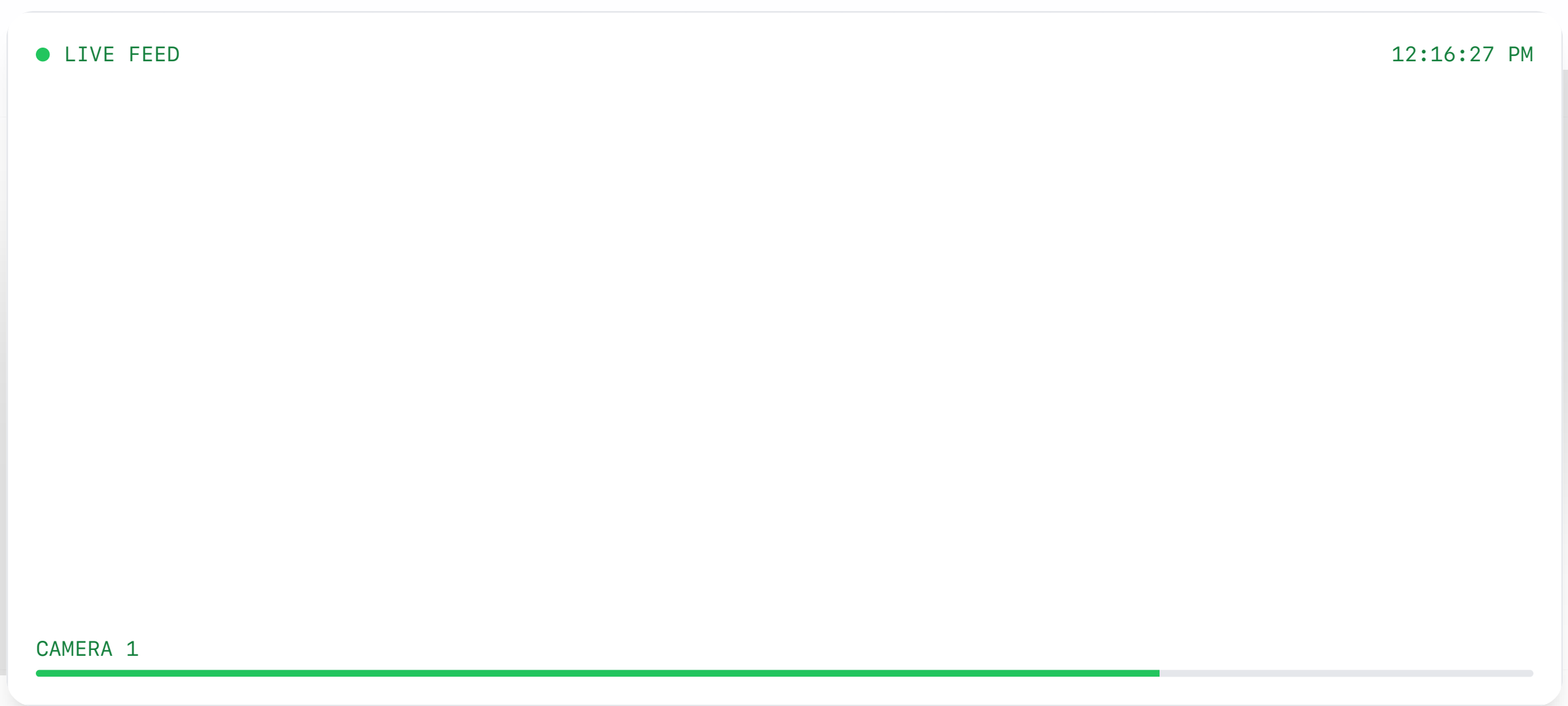




A cutting-edge solution that uses computer vision and geospatial intelligence to detect, report, and visualize violent incidents in real-time.



1 Live Camera Monitoring

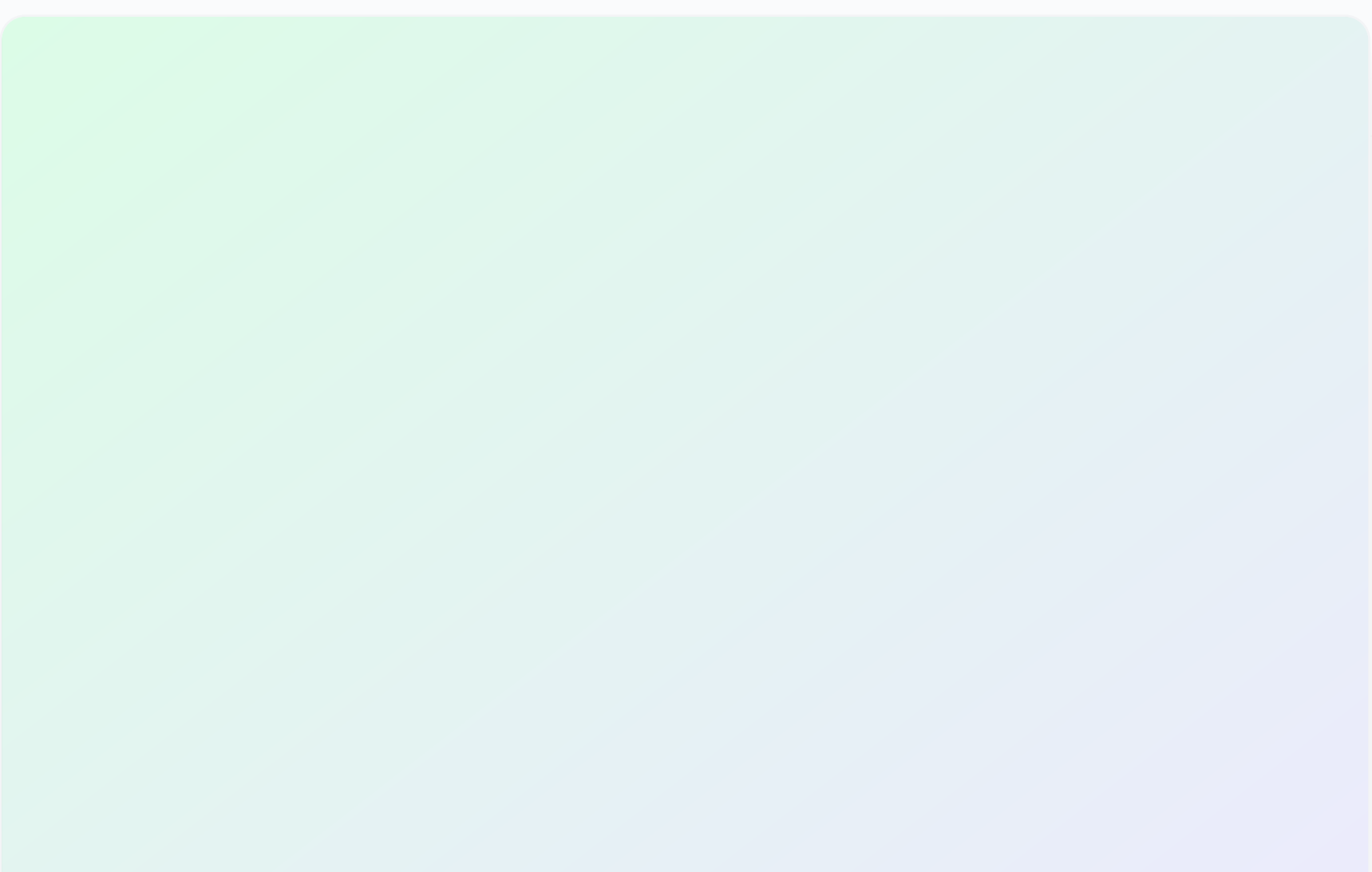
Users access a web interface displaying a live camera feed. The backend processes each video frame to detect violent actions in real-time.

2 Violence Detection Logic

Implemented using OpenCV, MediaPipe for human keypoints extraction, and YOLOv5 for detecting people in each frame.

3 Event Dashboard

A sleek, real-time dashboard displays events with image snapshots, map visualization, and filtering capabilities.



System Architecture
End-to-End Violence Detection Pipeline



November 2024

May 2025

April 2025

Final Implementation

April 2025

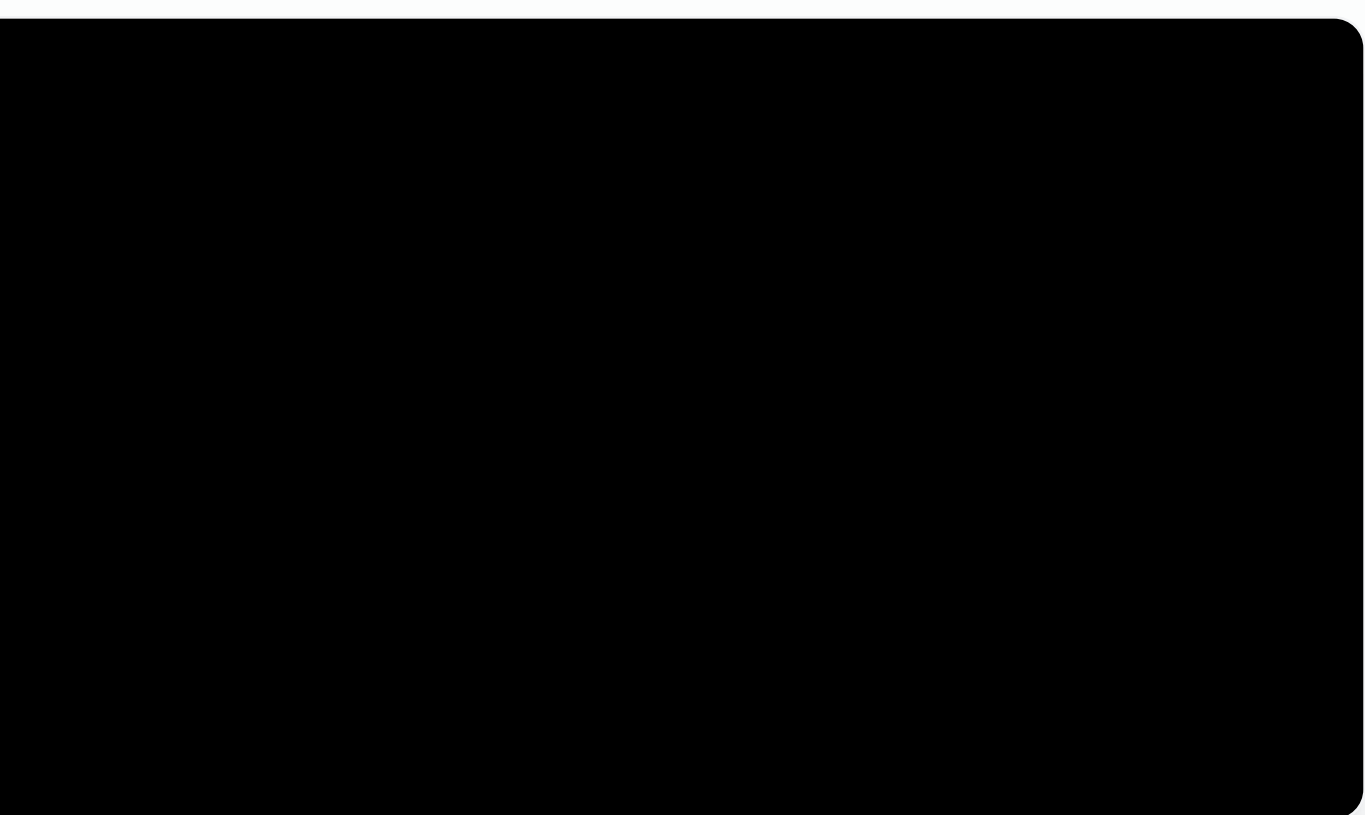
Complete system with YOLOv5 integration, geospatial intelligence, and fully functional dashboard.

98%

Detection Accuracy

<1s

Response Time



Real-time Detection

Powerful computer vision algorithms detect violent actions from live video feeds instantly.



Geospatial Intelligence

Precise location tracking with GPS and IP fallback for accurate incident mapping.



Instant Reporting

Automatic event detection with snapshot capture and immediate dashboard updates.

Tech Stack Overview

Live Video

OpenCV + Flask video stream

Detection

MediaPipe + YOLOv5 + NumPy

Backend

Python, Flask

Frontend UI

HTML, CSS, JavaScript

Event Transport

Node.js script to JSONBin

Location

HTML5 Geolocation + IP fallback

Storage

JSONBin.io (NoSQL)

Mapping

Leaflet.js



Contents of Event Dashboard

12 events today



42

Total Events



12

Today



28

Reviewed



8

Locations

All Events

Map View

Search events...

